

Partially Moment-Constrained Positive Stieltjes Minimax Quadrature

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Abstract

We consider fixed-budget quadrature for integrands containing a nearly singular centered Lorentzian component and an incompletely specified smooth residual. For n positive symmetric node pairs, a moment-depth parameter k , $0 \leq k \leq n$, imposes the first k moment conditions at infinity. The endpoints are an unrestricted positive Stieltjes rational minimax rule and a degree- $(2n - 1)$ -exact quadrature rule. Shared Laurent coefficients reduce the difference-numerator degree to at most $2n - k - 1$, yielding a $2n - k + 1$ -point alternation certificate within the stated admissible class. Complete high-precision hierarchies for $n = 2, 3, 4$ quantify the finite-budget price of polynomial protection. For smooth residuals with Chebyshev coefficients bounded by $Ch\omega^{-j}$, the exact worst-case smooth error is the weighted defect norm $C \sum_j h\omega^{-j} |I[T_j] - Q[T_j]|$. Adding the Lorentzian uncertainty gives an exact model-aware rule-selection risk. Interval arithmetic applied directly to the published rounded rules certifies a unique rule on 99.9703% of the tested parameter-domain area. A shifted-Lorentzian stress test and a separate verification code path delimit the scope and support reproducibility.

Keywords: numerical quadrature, rational approximation, Stieltjes functions, minimax approximation, validated numerics, uncertainty quantification

1. Introduction

Polynomial exactness is an effective design principle for smooth integrands, but it need not be the most efficient use of a fixed evaluation budget when the

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dominant difficulty is a nearby pole. Rationally exact quadrature incorporates prescribed singular structure [1], while generalized Gaussian constructions extend exactness to non-polynomial function spaces [2]. Optimization-based and positive cubature methods provide further ways to design nodes and weights [3, 4]. Related rational approximation theory for Markov functions explains the importance of negative real poles and positive residues [5].

The practically important intermediate case is partial trust: a structured nearly singular component is expected, but an unknown smooth background must also be protected. Exactness alone does not always predict practical quadrature performance [6]. We therefore introduce a hierarchy indexed by a moment depth k . Small k allocates more freedom to the structural family; large k provides stronger low-degree background protection.

The contribution is not a claim that constrained rational approximation or specialized quadrature is new in isolation. Interpolation-constrained rational minimax methods and common-support cubature provide close prior context [10, 7]. The proposed contribution is the combined framework of:

- (i) a positive partial moment-at-infinity hierarchy;
- (ii) a class-specific alternation count;
- (iii) an exact coefficient-space risk and rule selector;
- (iv) interval certification of the distributed rounded rules.

2. Model and admissible hierarchy

Let

$$I[f] = \int_{-1}^1 f(x) dx, \quad L_b(x) = \frac{1}{x^2 + b^2}, \quad b \in [0.08, 0.12].$$

With $s = b^2$ and $y = x^2$,

$$F(s) = I[L_{\sqrt{s}}] = \int_0^1 \frac{y^{-1/2}}{s + y} dy = \frac{2}{\sqrt{s}} \arctan \frac{1}{\sqrt{s}}.$$

A symmetric n -pair rule is

$$Q[f] = \sum_{j=1}^n w_j \{f(-x_j) + f(x_j)\}, \quad 0 < x_1 < \cdots < x_n < 1, \quad w_j > 0.$$

Writing $y_j = x_j^2$, its response on the structural family is

$$r_Q(s) = Q[L_{\sqrt{s}}] = 2 \sum_{j=1}^n \frac{w_j}{s + y_j}.$$

For $0 \leq k \leq n$, define $\mathcal{A}_{n,k}$ by

$$\sum_{j=1}^n w_j y_j^\ell = \frac{1}{2\ell + 1}, \quad \ell = 0, \dots, k-1.$$

Lemma 1 (Moment and polynomial exactness). *A symmetric rule belongs to $\mathcal{A}_{n,k}$ if and only if it is exact for every polynomial of degree at most $2k-1$.*

Proof. Odd polynomials are integrated exactly by symmetry. For an even monomial,

$$Q[x^{2\ell}] = 2 \sum_{j=1}^n w_j y_j^\ell, \quad I[x^{2\ell}] = \frac{2}{2\ell + 1}.$$

The stated equivalence follows. □

Lemma 2 (Laurent matching). *A rule belongs to $\mathcal{A}_{n,k}$ if and only if*

$$r_Q(s) - F(s) = O(s^{-k-1}), \quad s \rightarrow \infty.$$

Proof. Use the finite geometric expansion

$$\frac{1}{s + y} = \sum_{\ell=0}^{m-1} (-1)^\ell y^\ell s^{-\ell-1} + O(s^{-m-1}),$$

whose remainder is uniform for $y \in [0, 1]$. Integrating against $y^{-1/2} dy$ and summing against the discrete measure $2 \sum_j w_j \delta_{y_j}$ gives the two Laurent expansions. □

3. Degree reduction and alternation

Write $r_i = p_i/q_i$, where q_i is monic of degree n .

Proposition 3 (Difference-numerator degree). *If $r_1, r_2 \in \mathcal{A}_{n,k}$, then*

$$r_1 - r_2 = \frac{H}{q_1 q_2}, \quad \deg H \leq 2n - k - 1,$$

unless $H \equiv 0$.

Proof. The generic numerator $p_1q_2 - p_2q_1$ has degree at most $2n - 1$. Shared moment conditions give $r_1 - r_2 = O(s^{-k-1})$. Since q_1q_2 is monic of degree $2n$, a nonzero numerator of degree d must satisfy $d - 2n \leq -k - 1$. $\square$

Define the relative structural error

$$\varepsilon_Q(s) = \frac{r_Q(s)}{F(s)} - 1.$$

Theorem 4 (Alternation certificate). *Suppose $Q^* \in \mathcal{A}_{n,k}$ has $2n - k + 1$ ordered points at which ε_{Q^*} has equal magnitude and alternating sign, and this magnitude equals its uniform norm. Then no $Q \in \mathcal{A}_{n,k}$ has strictly smaller uniform relative error.*

Proof. A strictly better competitor would make $\varepsilon_{Q^*} - \varepsilon_Q$ alternate sign at all $2n - k + 1$ points, forcing at least $2n - k$ distinct zeros. Because $F(s) > 0$, these are zeros of $r_{Q^*} - r_Q$, contradicting the degree bound unless the rational responses are identical. $\square$

Remark 5. The theorem establishes minimaxity within the stated class. It does not by itself establish unconditional uniqueness or optimality over adaptive algorithms or different node budgets.

4. Numerical hierarchy

Complete high-precision hierarchies were computed for $n = 2, 3, 4$. Table 1 reports the final $n = 4$ family.

Table 1: The $n = 4$ partial-moment hierarchy.

k	Exactness degree	Alternation points	Worst relative error	Minimum w_j
0	-1	9	1.490623e-10	0.039147
1	1	8	2.460626e-09	0.044528
2	3	7	4.048641e-08	0.051670
3	5	6	6.653546e-07	0.061646
4	7	5	1.092819e-05	0.076595

Each added moment condition increases the structural error by roughly a factor of 16 in the computed range. This is an empirical finite-range observation, not an asymptotic theorem.

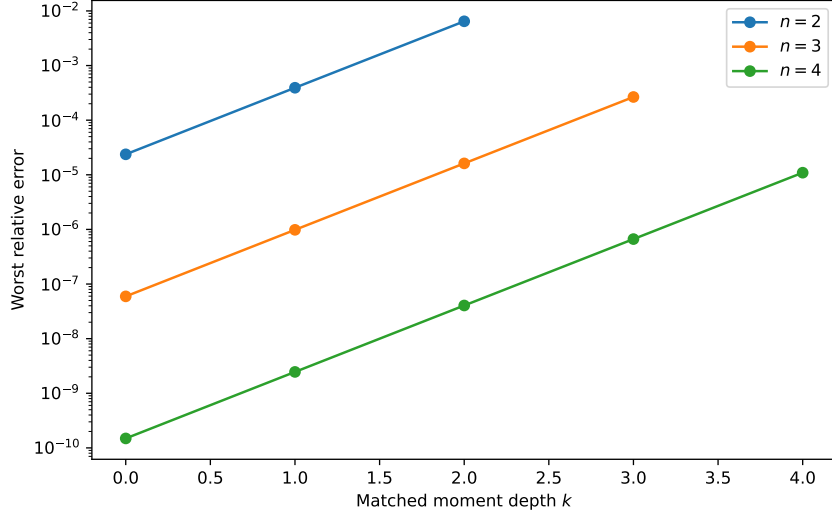


Figure 1: Worst relative error across the complete hierarchies for $n = 2, 3, 4$. Increasing k protects more moments but raises the target-family minimax error.

5. Exact coefficient-space risk

Let

$$h(x) = \sum_{j=0}^{\infty} a_j T_j(x), \quad |a_j| \leq C \rho^{-j}, \quad \rho > 1,$$

and define

$$\Delta_j(Q) = I[T_j] - Q[T_j].$$

Theorem 6 (Exact coefficient-box norm). *For every fixed bounded quadrature rule Q ,*

$$\sup_h |I[h] - Q[h]| = CG_Q(\rho), \quad G_Q(\rho) = \sum_{j=0}^{\infty} \rho^{-j} |\Delta_j(Q)|.$$

Proof. Uniform convergence permits termwise application:

$$I[h] - Q[h] = \sum_{j=0}^{\infty} a_j \Delta_j(Q).$$

The triangle inequality gives the upper bound. It is attained by $a_j = C \rho^{-j} \text{sign } \Delta_j(Q)$. $\square$

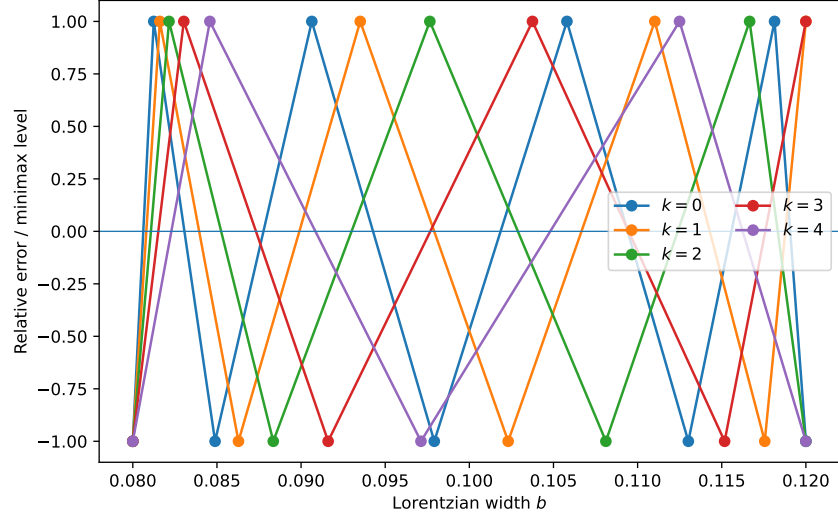


Figure 2: Normalized extremal errors for the $n = 4$ rules. The counts 9, 8, 7, 6, 5 agree with $2n - k + 1$.

Define

$$S_Q = \max_{b \in [0.08, 0.12]} |I[L_b] - Q[L_b]|.$$

Theorem 7 (Exact mixed-class risk). *For $f = h + cL_b$, with h in the coefficient box and $|c| \leq c_{\max}$,*

$$\sup_f |I[f] - Q[f]| = CG_Q(\rho) + c_{\max} S_Q.$$

The signs of the smooth and structural extremizers can be chosen independently. With $\tau = c_{\max}/C$, the robust choice minimizes $G_Q(\rho) + \tau S_Q$.

Table 2: Representative structural and smooth-risk components at $\rho = 1.5$.

k	Structural upper bound	$G_k(1.5)$	Generic bound / exact norm
0	5.629812e-09	9.112419e-01	11.47
1	9.179142e-08	3.954654e-01	13.49
2	1.509170e-06	2.046570e-01	11.58
3	2.480061e-05	9.598564e-02	10.98
4	4.073393e-04	4.492424e-02	10.42

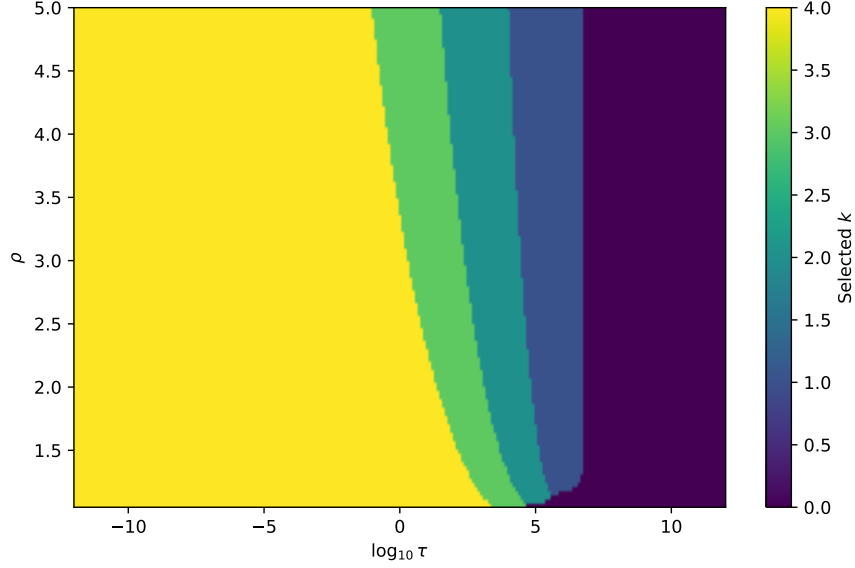


Figure 3: Moment depth selected by minimizing $G_k(\rho) + \tau S_k$.

6. Certification of the rounded rules

The ideal minimax rules satisfy the moments exactly. The distributed decimal rules have tiny but nonzero rounding defects. The coefficient-space risk remains exact for every fixed rounded operator, but the ideal degree-reduction theorem is not inherited from a nominal k -label alone.

We therefore certified the rounded rules directly. Structural maxima were enclosed over the complete b -interval, Chebyshev defects were enclosed through degree 500, the remaining tail was bounded analytically, and a parameter cell was labeled only when one risk upper bound was below every competing lower bound throughout the cell.

The unresolved area is 0.0297%, localized in thin cells intersecting transition curves rather than broad unvalidated regions.

7. External-validity stress test

Without retraining, the centered rules were tested on

$$L_{a,b}(x) = \frac{1}{(x-a)^2 + b^2}, \quad |a| \leq 0.03, \quad 0.08 \leq b \leq 0.12.$$

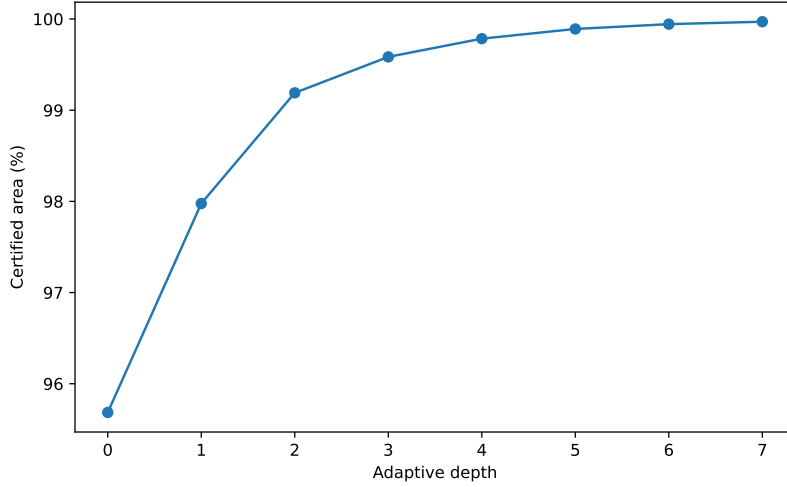


Figure 4: Growth of certified selector coverage under adaptive refinement. Final coverage is 99.9703%.

Table 3: Worst relative error on the shifted-Lorentzian stress-test box.

Method	Evaluations	Worst error	Worst $ a $	Worst b
PMC-k0	8	4.522336e-06	0.0300	0.0800
PMC-k1	8	1.804608e-05	0.0300	0.0800
PMC-k2	8	3.758665e-05	0.0300	0.0893
PMC-k3	8	3.918636e-04	0.0300	0.0800
PMC-k4	8	1.025564e-03	0.0300	0.0845
Gauss-Legendre-8	8	4.300499e-01	0.0000	0.0800
Composite-Simpson-9	9	7.426762e-02	0.0300	0.1200

The best existing hierarchy rule is PMC- $k = 0$, with worst relative error $4.522336e - 06$. This is approximately 95094.6 times smaller than the worst eight-point Gauss-Legendre error and 16422.4 times smaller than the nine-evaluation composite Simpson error on the same box. The result supports local robustness, not shifted-family minimaxity or shift invariance.

8. Independent verification and reproducibility

A separate verifier reconstructs moments, alternation points, centered errors, structural errors, transfer functions, and deterministic selector checks

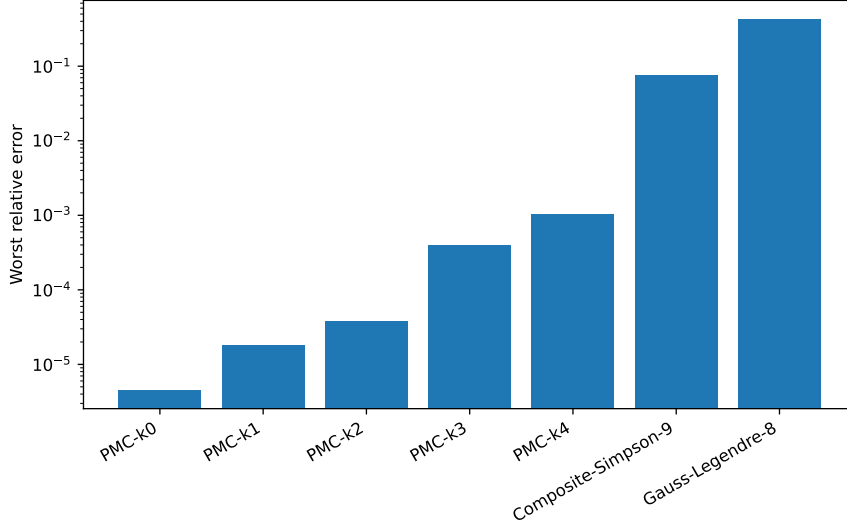


Figure 5: External-validity comparison without retraining.

from only the published nodes, weights, and formulas. It does not import the nonlinear optimization or interval-certification code.

All geometry and alternation checks passed. The largest constrained moment residual of the rounded rules was 5.551×10^{-17} . The maximum relative discrepancy in the transfer-function audit was 2.731×10^{-12} , and all scored certified selector spots agreed.

A one-command repository regenerates the principal tables and figures and runs seven automated tests. This is an internal independent reimplementaion, not external third-party replication.

9. Relation to prior work and contribution scope

Simultaneous and shared-support quadrature constructions provide additional context [8, 7]. Recent lower-bound work also emphasizes that upper-error estimates should be interpreted relative to the function class [9]. Our claim is limited to the combination of the partial moment-at-infinity hierarchy, its alternation structure, the exact coefficient-space selector, and direct certification of the rounded rules.

No claim of universal superiority, automatic model detection, formal proof-assistant verification, or a proved asymptotic product law is made.

10. Limitations

The main structural family is centered and one-dimensional. The shifted test covers only small location mismatch and is not interval-certified. The empirical separable error law is supplementary rather than theoretical. A third-party implementation and specialist novelty review remain desirable.

11. Conclusion

Partial moment constraints provide a controlled bridge between a specialized positive Stieltjes minimax rule and polynomial-exact quadrature. The hierarchy quantifies the price of protecting an unknown smooth background. Weighted Chebyshev coefficient boxes convert the trade-off into an exact robust risk and a model-aware selector. Interval computation certifies the rounded rules over nearly the entire tested domain.

Data and code availability

The rule tables, verification scripts, generated data, and figure reproduction code are available in the accompanying archive and are prepared for public release at <https://github.com/TurkeyOp/partial-moment-stieltjes-quadrature>. The repository should be made public before journal submission.

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Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to assist with literature organization, code prototyping, manuscript structuring, and language revision. All mathematical derivations, numerical results, references, interpretations, and final wording were reviewed and verified by the author, who takes full responsibility for the content of the article.

References

- [1] W. Gautschi, Gauss-type quadrature rules for rational functions, arXiv:math/9307223 (1993).
- [2] D. Huybrechs, On the computation of Gaussian quadrature rules for Chebyshev sets of linearly independent functions, arXiv:1710.11244 (2017).
- [3] V. Keshavarzzadeh, R. M. Kirby, A. Narayan, Generation of nested quadrature rules for generic weight functions via numerical optimization: Application to sparse grids, arXiv:1808.03707 (2018).
- [4] J. Glaubitz, Constructing positive interpolatory cubature formulas, arXiv:2009.11981 (2020).
- [5] B. Beckermann, J. Bisch, R. Luce, On the rational approximation of Markov functions, with applications to the computation of Markov functions of Toeplitz matrices, arXiv:2106.05098 (2021).
- [6] L. N. Trefethen, Exactness of quadrature formulas, arXiv:2101.09501 (2021).
- [7] J. R. Bravo, J. A. Hernández, S. Ares de Parga, R. Rossi, A subspace-adaptive weights cubature method with application to the local hyperreduction of parameterized finite element models, arXiv:2310.15769 (2023).
- [8] W. Van Assche, A Golub–Welsch version for simultaneous Gaussian quadrature, arXiv:2309.11864 (2023).

- [9] T. Goda, Y. Kazashi, K. Tanaka, How sharp are error bounds? Lower bounds on quadrature worst-case errors for analytic functions, arXiv:2401.07196 (2024).
- [10] L.-H. Zhang, Y.-N. Zhang, b-d-Lawson: a method for the interpolation constrained rational minimax approximation, arXiv:2502.10665 (2025).